



Contributions of National Key Forestry Ecology Projects to the forest vegetation carbon storage in China

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ABSTRACT

Planted forests in China have continuously stored large amounts of vegetation carbon since the implementation of the National Key Forestry Ecology Projects (NKFEs). However, the time variation of the contribution of NKFE-covered planted forests to the national forest vegetation carbon storage has not been systematically documented. Moreover, predicting the future vegetation carbon storage of NKFE-covered planted forests is of great significance for global climate change. Thus, we collected vegetation carbon density data on China's forests from 251 published articles to clarify the contributions and vegetation carbon sequestration potential of NKFE-covered planted forests. The results showed that the vegetation carbon density of NKFE-covered planted forests ranged from 23.49 to 68.88 Mg ha⁻¹, that slight differences existed among the NKFE-covered planted forests during 2009–2018, and that slight changes occurred from 1989 to 2018. The total contribution rate of NKFE-covered planted forests to China's forest vegetation carbon storage was 36.29% during 2009–2018, and the contribution rates of most NKFEs during 2009–2018 were higher than those during 1999–2008. The total vegetation carbon sequestration potential ranged from 4.22 to 8.67 Pg under four scenarios, in which NKFE-covered planted forests contributed 40%, 60%, 80% and 100% of the vegetation carbon storage of natural forests in the corresponding NKFE regions, respectively. These results indicate that NKFE-covered planted forests significantly contribute to the vegetation carbon storage of China's forests, and that NKFE-covered planted forests have large vegetation carbon sequestration potential, which is beneficial for future national and global forest carbon fixation.

1. Introduction

China has contributed 25% of the global net increase in leaf area, of which 42% was from forests (Chen et al., 2019). Moreover, China's planted forests, whose area accounts for 27% of the global planted forest area (FAO, 2015), store large amounts of carbon in vegetation (Dixon et al., 1994; Schulze et al., 2000; Fang et al., 2001) and significantly contribute to global carbon storage (Winjum and Schroeder, 1997). As a consequence, these forests play a paramount role in mitigating rising atmospheric carbon dioxide (CO₂) concentrations, which have been worsened by human disturbances. For instance, Fang et al. (2001, 2014b) assessed the vegetation carbon storage of planted forests in China from 0.25 Pg to 0.95 Pg from the 1970s to the 2000s, and showed that China's planted forests have contributed to reduced CO₂ emissions.

To protect natural forest resources, restore degraded ecosystems and

safeguard national ecological security, a large number of forests have been planted since the implementation of National Key Forestry Ecology Projects (NKFEs) beginning in the 1970s (Van Den Hoek et al., 2015; Zhang et al., 2000), and the total afforestation area of NKFEs accounts for 43.15% of China's forests (State Forestry Administration, 1995–2017). The NKFEs mainly include the Yangtze River and Zhujiang River Shelter Forest Projects (River Shelter Forest Project, RSFP), Beijing-Tianjin Sand Source Control Project (BTSSCP), Three-North Shelter Forest Program (TNSFP), Natural Forest Protection Project (NFPP) and Grain for Green Program (GGP), and detailed information concerning these NKFEs can be found in Supplementary Appx. 1. Previous studies suggested that the implementation of NKFEs has improved ecosystem services (such as soil retention, water retention and flood mitigation) (Deng et al., 2012; Ouyang et al., 2016; Wu et al., 2019), increased the area of China's forests (Liu et al., 2013; Wu et al., 2019) and changed the zonal pattern of China's forest litter production

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(You et al., 2017). Moreover, NKFEPS have contributed to the national forest carbon stock (Dixon et al., 1994; Fang et al., 2001; Pan et al., 2011; Fang et al., 2014b). For instance, the GGP (Zhang et al., 2010) and the BTSSCP (Liu et al., 2018) have significantly increased forest ecosystem carbon stocks. Lu et al. (2018) showed that carbon sequestration induced by national ecological restoration projects was 0.77 Pg C, accounting for over half of the total ecosystem carbon sequestration in the national ecological restoration project regions from 2001 to 2010. However, the time variations of the contributions of NKFEPCovered planted forests to the national forest vegetation carbon storage has not been systematically documented, which is important for support management and policy decisions.

Predicting the future vegetation carbon storage of NKFEPCovered planted forests is of great significance for global climate change, because NKFEPCovered planted forests significantly contribute to global carbon storage (Dixon et al., 1994; Fang et al., 2001). Moreover, identifying the crucial environmental factors and their relationships with the vegetation carbon density of NKFEPCovered planted forests is essential for predicting the future evolution of NKFEPCovered forest vegetation carbon storage. It is well known that climatic factors (e.g., precipitation and temperature) (Liu et al., 2018; Zhang et al., 2016), stand factors (e.g., stand age) (Cheng et al., 2015; Sun et al., 2016) and anthropogenic factors (e.g., government policies and artificial disturbances) (Ouyang et al., 2016) have played key roles in altering the vegetation carbon density of NKFEPCovered forests. However, their effects likely differ in direction and magnitude among NKFEPCovered forests due to different limiting factors in different NKFEPCovered regions, and the critical factors of the NKFEPCovered forest vegetation carbon density have not yet been systematically explored.

In this study, we aimed to answer the following questions by collecting data from published articles. (1) How much did NKFEPCovered planted forests contribute to China's forest vegetation carbon storage during the past decade, and how did the contribution rates vary with time? (2) What were the contribution rates of different NKFEPCovered planted forests during the past decade? (3) What is the carbon sequestration potential of NKFEPCovered planted forests? The results are expected to provide scientific evidences for managing China's NKFEPCovered forests under the effects of global changes.

2. Materials and methods

2.1. Site description

China is a vast country with a complex geographical environment, ranging from tropical to boreal zones and from heavily agricultural lowlands to the world's highest plateau (the Qinghai-Tibetan Plateau). As a result, China has abundant forest biomes, which include tropical rainforests, tropical seasonal rainforests, temperate forests, boreal forests, etc. (Fang et al., 2013). Due to the large-scale implementation of NKFEPS, planted forests are widely distributed in China. The RSFP, BTSSCP and TNSFP were implemented in specific regions. The RSFP was mainly implemented in southwest, south and southeast China, including 17 provinces (e.g., Qinghai, Gansu, Sichuan, etc.); the implementation area of the BTSSCP included Inner Mongolia, Shanxi, Hebei, Beijing, and Tianjin Provinces; and the TNSFP was mainly carried out in northwest, north and northeast China, including 13 provinces (e.g., Xinjiang, Qinghai, Gansu, etc.). In contrast, the NFPP and GGP have been implemented throughout China. The NFPP has been mainly implemented in 17 provinces (e.g., Sichuan, Gansu, Shaanxi, etc.), and the GGP covers almost all provinces in China (Supplementary Fig. 1).

2.2. Data compilation

Published journal articles (1999–2018) were collected using the China National Knowledge and Web of Science databases, and the

collected data were compiled into a database. The remaining data from published journal articles were extracted from graphs using the Get Data Graph Digitizer (version 2.24, Russian Federation). To ensure comparability among different periods and to make international comparisons, only previous studies that satisfied the following criteria were included in the database for analysis: (1) the forests satisfied the definition of the FAO (2010), and selected forest types (mangroves, bamboo and rubber planted forests) and other wooded lands (shrubwood and urban forests) were excluded; (2) there was a clear experimental (or inventory) time for the data; (3) there was a clear forest origin (planted forest or natural forest) for the data; and (4) the data were collected under natural conditions, with data from other treatments excluded. In total, 251 published studies met the criteria (Supplementary Appx. 2), which comprised 1191 datapoints of planted forests and 893 datapoints of natural forests throughout China (Supplementary Fig. 2). In addition, the databases required in this study included the vegetation carbon density (Mg ha^{-1}) (i.e., the total vegetation carbon density and the carbon density of the overstory layer and understorey layer), geographical factors (i.e., latitude ($^{\circ}$), longitude ($^{\circ}$), altitude (m) and gradient ($^{\circ}$)), experimental (or inventory) time, climatic factors (i.e., mean annual temperature (MAT, $^{\circ}\text{C}$), mean annual precipitation (MAP, mm), mean annual accumulated temperature (MAAT, $^{\circ}\text{C}$), mean annual evaporation (MAE, mm) and humidity (%)), stand factors (i.e., forest stand origin (planted forest and natural forest), stand age (year), canopy density and stem density ($\text{individual ha}^{-1}$)) and edaphic factors (i.e., soil bulk density (g cm^{-3}) and soil depth (cm)).

2.3. Data analysis

The geographical distributions of the planted forests were identified according to information in the literature, and ArcGIS (version 10.5, USA) was used to determine the geographical distributions in different NKFEPCovered regions (Supplementary Fig. 1). Thus, the contribution rate of NKFEPCovered planted forests to the vegetation carbon storage of China's forests was calculated by the following formula (1):

$$\text{rate}_{ij} = (CD_{ij} \times \text{Area}_{ij}) / CS_j \quad (1)$$

where rate_{ij} is the contribution rate of NKFEPCovered planted forests to the carbon storage of China's forests during period j , CD_{ij} is the average vegetation carbon density of NKFEPCovered planted forests during period j , and Area_{ij} represents the existing afforestation area during period j , which was calculated as the summation of the annual new afforestation area for NKFEPCovered up to the last year of period j . The annual new afforestation area from 1978 to 2016 was taken from the China Forest Statistical Yearbook (State Forestry Administration, 1995–2017). In addition, the afforestation area for NKFEPCovered during the 2009–2018 period was calculated by summing the annual afforestation area up to 2016 because the afforestation area of NKFEPCovered from 2017 to 2018 was unavailable. CS_j is the total carbon storage of China's forests during period j . The value of CS during 1989–1998 was taken from the study of Fang et al. (2007), and the values of CS during 1999–2008 and 2009–2018 were referenced from the studies of Fang et al. (2014a), Tang et al. (2018), respectively.

Natural forests undergo a relatively stable state after a long period of succession; therefore, the vegetation carbon storage of natural forests in NKFEPCovered regions was considered the maximum vegetation carbon storage that the corresponding NKFEPCovered planted forests could achieve in the natural environment. Thus, the vegetation carbon sequestration potential of NKFEPCovered planted forests was defined as the net carbon sequestration increment on the basis of the existing vegetation carbon levels of NKFEPCovered planted forests due to the impacts or management of natural factors or human factors (formula 2):

$$\text{Potential}_i = (CD_{na-i} \times \text{Percentage} - CD_i) \times \text{Area}_i \quad (2)$$

where Potential_i is the carbon sequestration potential of planted forests

in an NKFE_P region, CD_{na-i} is the vegetation carbon density of natural forests in an NKFE_P region during 2009–2018, and *Percentage* is the percentage of natural forest carbon storage in an NKFE_P region that the NKFE_P-covered planted forests could achieve under different conditions. The minimum proportion of the vegetation carbon density of NKFE_P-covered planted forests relative to that of natural forests in the corresponding NKFE_P region was 29.20% during 2009–2018 (Supplementary Table 1). Therefore, we assumed that NKFE_P-covered planted forests could obtain 40%, 60%, 80% and 100% of the vegetation carbon storage of natural forests in an NKFE_P region in the future. CD_i is the vegetation carbon density of NKFE_P-covered planted forests during 2009–2018, and $Area_i$ is the existing afforestation area of NKFE_P during 2009–2018. The study estimated the future potentials of NKFE_P-covered planted forests as carbon sinks, assuming that the areas of the forests remained unchanged and that the Chinese government did not fell the forests after 2018. Thus, we assumed that the potential for carbon sequestration was zero if the value of Potential_{*i*} was negative.

2.4. Statistical analysis

Two-tailed one-way ANOVA and Student's *t*-tests were used to compare the different samples when the samples were homogeneous, and the forms of "1/*X*", " $X^{0.5}$ ", and "Ln (*X*)" were used to adjust the variance of the samples. A two-tailed nonparametric test was used to compare the different samples if the samples were not homogeneous. The application of these methods can be found in Supplementary Appx. 3. The classification and regression tree (CART) model was used to identify critical factors for the vegetation carbon density, and the details of CART operations can be found in the study of Sun et al. (2013). The minimum sample size for CART was 26. Details of the CART process can be found in Supplementary Figs. 3–12. In addition, regression analysis was performed to analyze the relationships between these critical factors and the vegetation carbon density, and a descriptive statistical analysis was used to understand the distribution and variability of the forest stand age with a density curve. The statistical tests were considered significant at the $P < 0.05$ level.

One-way ANOVA, Student's *t*-test, the nonparametric test and regression analyses were carried out with IBM SPSS statistical software (version 20.0, USA). The descriptive statistical analysis was carried out with R software (version 3.5.1, R Core Team), and the CART analysis was conducted in R software (version 3.5.1, R Core Team) using the "repart" package. The graphs were constructed using Sigma Plot (version 12.5, USA).

3. Results

3.1. Vegetation carbon density of planted forests in China

The average vegetation carbon densities of China's planted forests were 31.67, 46.03, and 52.91 Mg ha⁻¹ during 1989–1998, 1999–2008, and 2009–2018, respectively. Correspondingly, the average vegetation carbon densities of China's natural forests were 51.55, 124.84, and 92.79 Mg ha⁻¹ during 1989–1998, 1999–2008, and 2009–2018, respectively. Moreover, the vegetation carbon density of planted forests was always significantly ($P < 0.05$) lower than that of natural forests during each period. The vegetation carbon density of the planted forests changed slightly ($P > 0.05$) over time (Fig. 1a).

The average carbon densities of the overstory layer in the planted forests were 8.66, 11.82, 22.18, 35.13, and 41.66 Mg ha⁻¹ during 1969–1978, 1979–1988, 1989–1998, 1999–2008, and 2009–2018, respectively, and those in the natural forests were 24.10, 48.74, 53.32, 57.29, and 66.71 Mg ha⁻¹, respectively. Additionally, the carbon density of the overstory layer of the planted forests was significantly ($P < 0.05$) lower than that of the natural forests in each period, and the carbon densities of the overstory layer in the planted forests and in the natural forests significantly ($P < 0.05$) increased over time

(Fig. 1b).

The average carbon densities of the understory layer in the planted forests were 3.21, 2.84, and 2.10 Mg ha⁻¹ during 1989–1998, 1999–2008, and 2009–2018, respectively, and those in the natural forests were 1.73, 5.15, and 4.82 Mg ha⁻¹, respectively. Slight ($P > 0.05$) differences were found between the planted forests and natural forests during 1989–1998 and 1999–2008, but the carbon density of the understory layer in the natural forests was significantly ($P < 0.05$) higher than that in the planted forests during 2009–2018. Moreover, the carbon density of the understory layer in the planted forests significantly ($P < 0.05$) decreased over time, and that in the natural forests changed slightly over time ($P > 0.05$) (Fig. 1c).

3.2. Vegetation carbon densities of the planted forests in different NKFE_P regions

The average vegetation carbon densities ranged from a low of 23.49 Mg ha⁻¹ for BTSSCP-covered planted forests to a high of 68.88 Mg ha⁻¹ for RSFP-covered planted forests during 2009–2018, and no significant ($P > 0.05$) differences in the vegetation carbon density of planted forests were found among the different NKFE_P regions during 2009–2018 (Fig. 2). Moreover, significant variations in the vegetation carbon density of NKFE_P-covered planted forests were not observed over time (Fig. 3a). The average carbon densities of the overstory layer were 42.79, 26.54, 29.47, 35.47, and 39.81 Mg ha⁻¹ for the planted forests in the RSFP, BTSSCP, TNSFP, NFPP and GGP regions during 2009–2018, respectively, and the carbon density of the overstory layer in the TNSFP-covered planted forests was significantly ($P < 0.05$) lower than that in the other NKFE_P-covered planted forests (Fig. 2). Additionally, the carbon density of the overstory layer of all NKFE_P-covered planted forests changed slightly ($P > 0.05$) over time, with the exception of that of the TNSFP-covered planted forests (Fig. 3b). The average carbon density of the understory layer ranged from a low of 1.03 Mg ha⁻¹ for the TNSFP-covered planted forests to a high of 2.79 Mg ha⁻¹ for the RSFP-covered planted forests during 2009–2018, there were no significant differences in the carbon densities of the understory layer among the different NKFE_P-covered planted forests (Fig. 2), and the carbon densities of the understory layer of all NKFE_P-covered planted forests changed slightly ($P > 0.05$) over time (Fig. 3c).

3.3. Contributions and carbon sequestration potentials of NKFE_P-covered planted forests

The contribution rates of planted forests to the vegetation carbon storage of China's forests were 5.30%, 1.54%, 10.93%, 4.81%, and 13.71% in the RSFP, BTSSCP, TNSFP, NFPP, and GGP regions during 2009–2018, respectively. The total contribution rates of NKFE_P-covered planted forests to the national forest vegetation carbon storage were 24.77%, 43.52%, and 36.29% during 1989–1998, 1999–2008, and 2009–2018, respectively. Moreover, the contribution rates of most NKFE_Ps during 2009–2018 were higher than those during 1999–2008, but the contribution rate of the TNSFP during 2009–2018 was lower than that during 1999–2008 (Fig. 4a).

The NKFE_P-covered planted forests had different carbon sequestration potentials. The maximum carbon sequestration potential was found in the TNSFP region, and the minimum potential occurred in the RSFP region. The total carbon sequestration potentials of NKFE_P-covered planted forests were 0.44 Pg, 1.62 Pg, 3.24 Pg, and 4.89 Pg under the scenarios of 40%, 60%, 80%, and 100% of the natural forest carbon storage, respectively (Fig. 4b).

3.4. Effects of biotic and abiotic factors on the vegetation carbon densities of NKFE_P-covered planted forests

The stand age was a critical factor determining the vegetation

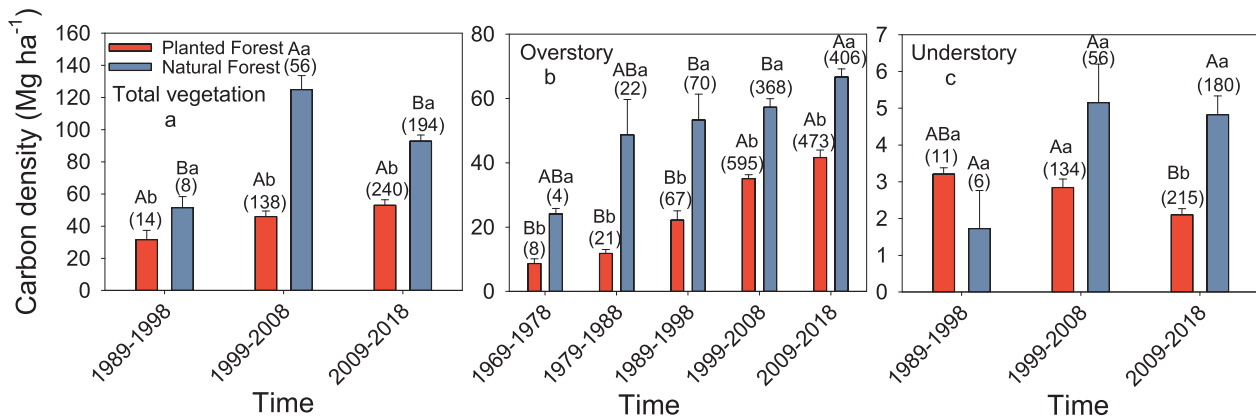


Fig. 1. Dynamics of the vegetation carbon densities of different forests with different stand origins. Different capital letters represent significant variations in natural forests or planted forests over time at the 0.05 level. Different lowercase letters indicate significant differences between natural forests and planted forests during the same period at the 0.05 level. The values are the means $\pm$ the SE, and the sample size numbers are shown in parentheses.

carbon density of most of the NKFP-covered planted forests (Table 1, Supplementary Figs. 13–17). The vegetation carbon densities of all NKFP-covered planted forests significantly ($P < 0.05$) increased with the stand age, and the coefficients between the stand age and the vegetation carbon densities of most of the NKFP-covered planted forests were higher than those between the stand age and the vegetation carbon densities of natural forests in the corresponding NKFP regions (Supplementary Fig. 18b–f). Topographic factors (longitude, latitude, altitude, and slope) were critical factors (Table 1, Supplementary Figs. 13–17), and the MAT and MAP were critical factors among the meteorological factors. The MAT was a critical factor affecting the vegetation carbon densities of NFPP-covered planted forests, and the MAP was a critical factor affecting the vegetation carbon densities of planted forests in both the NFPP and GGP regions (Table 1, Supplementary Figs. 15–17). Significantly ($P < 0.05$) positive relationships were observed between the MAP and the vegetation carbon densities of planted forests in the NFPP and GGP regions (Supplementary Fig. 19i, j). Moreover, the canopy density was a critical factor affecting the vegetation carbon densities of planted forests in both the BTSSCP and TNSFP

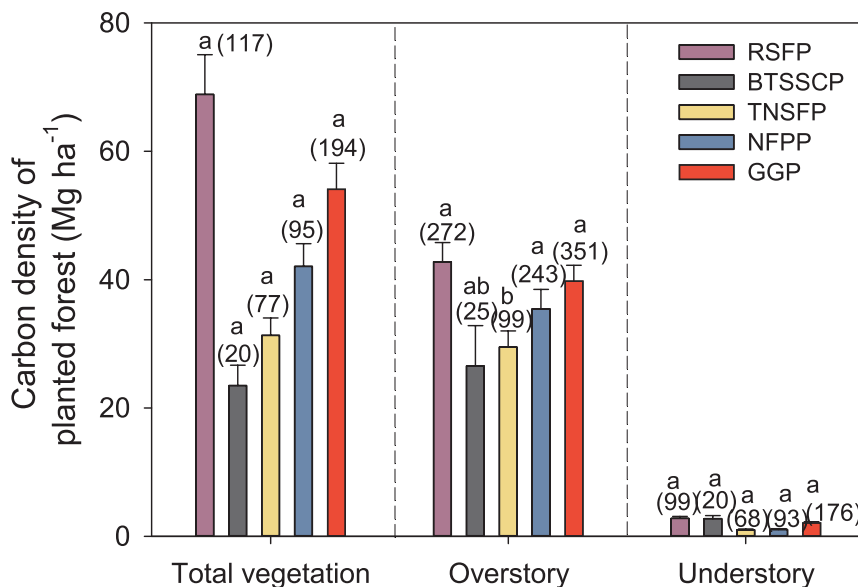
regions, and the stem density was a critical factor affecting the vegetation carbon densities of NFPP-covered planted forests (Table 1, Supplementary Figs. 14–16). The vegetation carbon density of TNSFP-covered planted forests was significantly ($P < 0.05$) positively correlated with the canopy density (Supplementary Figs. 20c).

4. Discussion

4.1. Vegetation carbon density of planted forests in China

The average vegetation carbon density of planted forests in China was 52.91 Mg ha^{-1} during 2009–2018 (Fig. 1), which is similar to the latest estimate of the forest biomass carbon density in China (55.7 Mg ha^{-1}) (Tang et al., 2018). However, the average carbon density of the overstory layer in the planted forests (41.66 Mg ha^{-1}) was higher than the value estimated based on Forest Inventory Data (FID) from the latest Chinese national survey during 2009–2013 (27 Mg ha^{-1}) (Shao et al., 2017). The main reason may be that the latter estimated value excluded 14.02 Mg ha^{-1} for China's planted

Fig. 2. Vegetation carbon densities of the planted forests in different NKFP regions during 2009–2018. RSFP: the River Shelter Forest Project. BTSSCP: the Beijing-Tianjin Sand Source Control Project. TNSFP: the Three-North Shelter Forest Program. NFPP: the Natural Forest Protection Project. GGP: the Grain for Green Program. Different letters represent significant differences in vegetation carbon density in the same layer among different regions at the 0.05 level. The values are the means $\pm$ the SE, and the sample size numbers are shown in parentheses.



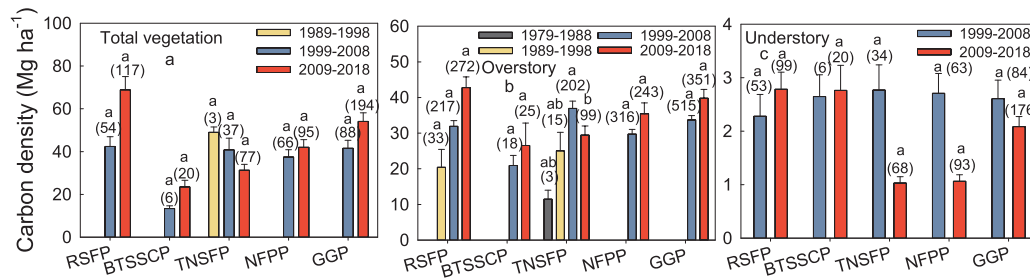


Fig. 3. The dynamics of the carbon densities of NKFEF-covered planted forests in China. RSFP: the River Shelter Forest Project. BTSSCP: the Beijing-Tianjin Sand Source Control Project. TNSFP: the Three-North Shelter Forest Program. NFPP: the Natural Forest Protection Project. GGP: the Grain for Green Program. Different letters represent significant variations in vegetation carbon density of the same NKFEF-covered planted forest over time at the 0.05 level. The values are the means $\pm$ the SE, and the sample size numbers are shown in parentheses.

forest root systems, which is based on an estimate of the root biomass of China's planted forests (28.04 Mg ha⁻¹) (Zhang et al., 2015).

The average carbon density of the overstory layer and the total vegetation in the planted forests were significantly ($P < 0.05$) lower than those in the natural forests in each period (Fig. 1), which is consistent with previous studies (Fang et al., 2001; Guo et al., 2013). These results could be because the planted forests were younger than the natural forests, and the vegetation carbon density significantly increased with the stand age (Supplementary Fig. 18a, Supplementary Fig. 21).

Our results showed that the carbon density of the understory vegetation in China's planted forests was generally lower than that in the natural forests. The reason may be that planted forests have monospecific, even-sized canopy structures with homogeneous tree sizes and spatial distributions, which result in more competition for light, water and nutrients. As a result, the growth of understory vegetation in the planted forests was limited (Bailey and Tappeiner, 1998; Hernandez-Tecles et al., 2015). However, we found that the carbon density of understory vegetation in the planted forests was not significantly ($P > 0.05$) different from that in the natural forests in the early stage. This result may have occurred because arbor species in the planted forests gradually became dominant in the ecosystem over time and gradually inhibited the growth of understory vegetation due to competition for light and belowground resources (Alaback, 1982; Jules

et al., 2008). Additionally, our results also showed that the carbon density in the understory layer of planted forests significantly decreased over time (Fig. 1c), which confirmed this idea.

4.2. Vegetation carbon densities of the planted forests in different NKFEF regions

No significant ($P > 0.05$) differences in the vegetation carbon densities of planted forests were found among the different NKFEF regions during 2009–2018. Various biotic and abiotic factors affected the vegetation carbon densities of NKFEF-covered planted forests (Table 1), and the effects of different factors on the vegetation carbon densities of NKFEF-covered planted forests differed in direction and intensity (Supplementary Figs. 18–20). As a result, the net interactive effect of environmental factors on the vegetation carbon density of NKFEF-covered planted forests was insignificant. Thus, no significant differences in the vegetation carbon densities of NKFEF-covered planted forests were found. For example, the climatic conditions in the RSFP region were the best among the five NKFEF regions, and these conditions were suitable for the growth of vegetation. However, the youngest RSFP-covered planted forests accumulate smaller amounts of carbon among the five NKFEF-covered planted forests, and the opposite is true of BTSSCP-covered planted forests (Supplementary Table 2).

Our results showed that the carbon density of the overstory layer in

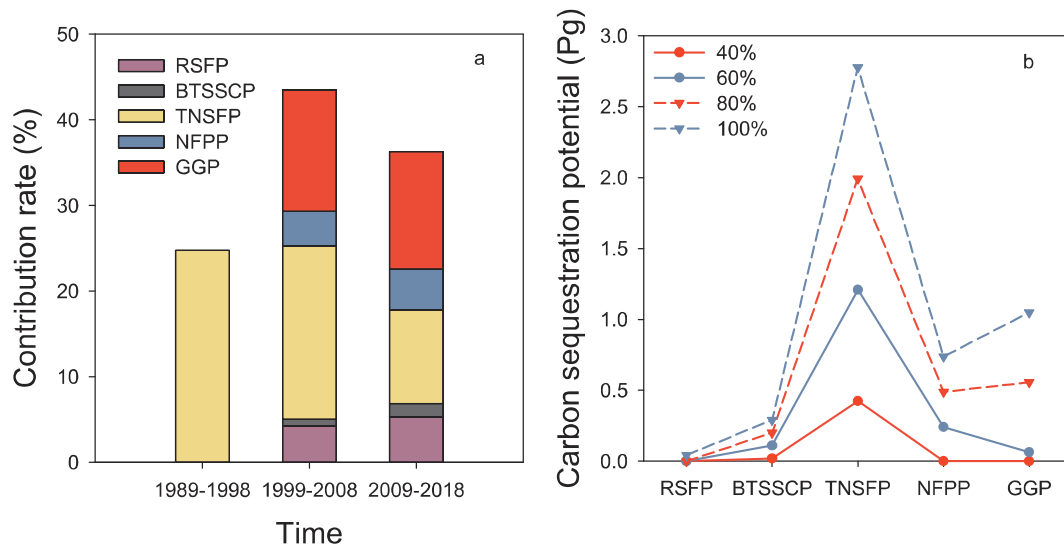


Fig. 4. Contribution rates (%) of the NKFEF-covered planted forests to the vegetation carbon storage of China's forests (a) and the carbon sequestration potentials of NKFEF-covered planted forests (b) under four scenarios (40%, 60%, 80% and 100% of natural forest carbon storage). RSFP: the River Shelter Forest Project. BTSSCP: the Beijing-Tianjin Sand Source Control Project. TNSFP: the Three-North Shelter Forest Program. NFPP: the Natural Forest Protection Project. GGP: the Grain for Green Program.

Table 1

Critical environmental factors of the vegetation carbon density of planted forests in NKFEF regions.

	RSFP	BTSSCP	TNSFP	NFPF	GGP
Topographic factor		Longitude Latitude Altitude	Altitude	Longitude Altitude Latitude Slope	Altitude Latitude Slope
Meteorological factor				MAT MAP	MAP
Stand factor	Stand age	Stand age Canopy density	Canopy density	Stand age Stem density	Stand age

Note: RSFP, BTSSCP, TNSFP, NFPF, and GGP represent the River Shelter Forest Project, Beijing-Tianjin Sand Source Control Project, Three-North Shelter Forest Program, Natural Forest Protection Project, and Grain for Green Program, respectively. MAP and MAT represent the mean annual precipitation and mean annual temperature, respectively.

the TNSFP-covered planted forests was significantly ($P < 0.05$) lower than those in the other NKFEF regions, which is likely due to water constraints. There is little precipitation in the TNSFP region (Supplementary Table 2), which is an arid and semi-arid area (Iglesias et al., 2012). As a result, the growth of vegetation was hindered. Moreover, *Populus* spp. that has been planted widely in the TNSFP region is independent of precipitation (Yan et al., 2011), resulting in the water availability decreasing with *Populus* spp. growth, and there is likely associated negative feedback to plant growth. As a result, the vegetation carbon density of the overstory layer in TNSFP-covered planted forests was lower than those in the other NKFEF-covered planted forests. Moreover, our results showed that the canopy density was a critical factor, being significantly ($P < 0.05$) positively correlated with the vegetation carbon density of TNSFP-covered planted forests (Table 1, Supplementary Fig. 20c). This result indicates that the vegetation carbon density of TNSFP-covered planted forests could be increased by artificially altering the canopy structure.

4.3. Contributions of NKFEF-covered planted forests to the forest vegetation carbon density in China

The contribution rate of GGP-covered planted forests to the carbon storage of China's forests was the highest, and that of BTSSCP-covered planted forests was the smallest among the five NKFEF-covered planted forests during 2009–2018 (Fig. 4a). There are two reasons that explain this phenomenon. First, the area of planted forests directly affects the contributions of NKFEFs. According to the China Forest Statistical Yearbook (State Forestry Administration, 1995–2017), the area of the GGP-covered planted forests was second largest after that of the TNSFP-covered planted forests, and that of the BTSSCP-covered planted forests was the smallest among the five NKFEF-covered planted forests. Second, the climatic environments in NKFEF regions may influence the contribution rates by altering the vegetation carbon densities of NKFEF-covered planted forests. A relatively better climatic environment in the GGP region promoted carbon accumulation in the GGP-covered planted forests, and the opposite was true of the TNSFP-covered planted forests (Supplementary Table 2).

Our results showed that the total contribution rates of NKFEF-covered planted forests during 2009–2018 were lower than those during 1999–2008, and that was mainly related to the contribution rate of the TNSFP-covered planted forests during 2009–2018 being lower than that during 1999–2008 (Fig. 4a). The TNSFP-covered planted forests grew stably more than 20 years after implementation of the TNSFP, which resulted in slight changes of the vegetation carbon density of TNSFP-covered planted forests (Fig. 3a). However, the vegetation carbon storage of China's forests has increased rapidly in recent decades due to

the changes in forest areas and age structures (Tang et al., 2018). Thus, the above two reasons led to a reduction in the contribution rate of the TNSFP-covered planted forests to the vegetation carbon storage of China's forests.

4.4. Future evolution of the contribution and vegetation carbon storage of NKFEF-covered planted forests

The total carbon sequestration potential of NKFEF-covered planted forests varied from 0.44 Pg to 4.89 Pg under the four scenarios (Fig. 4b), indicating that the vegetation carbon storage of NKFEF-covered planted forests will increase in the future. As a critical factor for the vegetation carbon density of most NKFEF-covered planted forests, the stand age of NKFEF-covered planted forests was very young (Table 1, Supplementary Table 2), and many dominant species in the planted forests (e.g., *Larix olgensis*, *Cyclobalanopsis glauca*, *Pinus tabulaeformis*, etc.) will continue to grow in the future (Cheng et al., 2014; Zhang et al., 2014; Zhao et al., 2011). In addition, our results showed that the vegetation carbon density of NKFEF-covered planted forests significantly ($P < 0.05$) increased with the stand age, and the coefficients between the stand age and the vegetation carbon densities of most of the NKFEF-covered planted forests were higher than those between the stand age and the vegetation carbon densities of natural forests in the corresponding NKFEF regions (Supplementary Fig. 18b–f). This finding indicates that the increasing rates of NKFEF-covered planted forests will be higher than those of natural forests in the future, and the contributions of NKFEF-covered planted forests to China's forest vegetation carbon storage will increase.

Government policies further promote vegetation carbon accumulation in China's forests (Ouyang et al., 2016). Under the call of Chairman Xi, who said that “China should tighten reforestation, expand the area of forests, improve the quality of forests, and enhance the ecological function of forests”, the Chinese government has attached great importance to forest ecological construction and has established a forestry development strategy that prioritizes ecological development. Thus, we believe that NKFEF-covered planted forests will sequester more carbon with improvements in China's planted forest quality and increases in planted forest area in the future.

The future climatic environment may promote carbon accumulation in NKFEF-covered planted forests. Our study showed that the vegetation carbon density of GGP-covered planted forests was positively correlated with the MAT (Supplementary Fig. 19e), indicating that NKFEF-covered planted forests will sequester more carbon due to global warming. Moreover, previous studies suggested that global warming (Fu et al., 2015; Peng et al., 2009) and rapid increases in nitrogen deposition (Lu et al., 2012; Schulte-Uebbing and de Vries, 2017) and atmospheric CO₂ concentrations (Peng et al., 2009; Yao et al., 2018) due to intensified anthropogenic activity have promoted vegetation carbon accumulation in China's planted forests. Overall, the contributions of NKFEF-covered planted forests to China's forest vegetation carbon storage will increase with the growth of planted forests, the further implementation of NKFEFs and global climate change.

Several uncertainties in our estimates of the vegetation carbon density of NKFEF-covered planted forests must be acknowledged. First, using average values to represent the vegetation carbon densities of NKFEF-covered planted forests may have caused bias, because planted forests in different study sites include different tree species, different climates and other growth characteristics. However, we collected as much forest vegetation carbon density data as possible from published papers to enrich the sample size, so that the average value would represent the vegetation carbon density. Second, the estimation of the vegetation carbon densities of NKFEF-covered planted forests ignores the vegetation carbon density caused by harvesting, which could result in an underestimation of the vegetation carbon density. However, the vegetation carbon density caused by harvesting could be small, because the harvesting amount of NKFEF-covered planted forests is very small

due to the restrictions of Chinese laws.

5. Conclusions

The total contribution rate of NKFEF-covered planted forests to China's forest vegetation carbon storage was 36.29% during 2009–2018, with contribution rates of 5.30%, 1.54%, 10.93%, 4.81%, and 13.71% in the RSFP-covered, BTSSCP-covered, TNSFP-covered, NFPP-covered, and GGP-covered planted forests, respectively. The contribution rates of most NKFEFs during 2009–2018 were higher than those during 1999–2008. The total carbon sequestration potential varied from 0.44 Pg to 4.89 Pg under the four scenarios. The vegetation carbon density of NKFEF-covered planted forests varied from 23.49 Mg ha⁻¹ to 68.88 Mg ha⁻¹, and slight differences in the vegetation carbon density of planted forests were found among the different NKFEF regions during 2009–2018. Moreover, slight variations in the vegetation carbon density of the NKFEF-covered planted forests over time were found. These findings suggest that NKFEFs significantly contribute to the vegetation carbon storage of China's forests, and that NKFEF-covered planted forests have a large vegetation carbon sequestration potential, which is beneficial for fixing more CO₂ in the future.

CRediT authorship contribution statement

Yu Zhang: Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing. **Ji Yuan:** Software, Visualization. **Chengming You:** Conceptualization, Methodology, Writing - original draft. **Rui Cao:** Investigation, Writing - original draft. **Bo Tan:** Project administration, Validation. **Han Li:** Validation, Writing - review & editing. **Wanqin Yang:** Conceptualization, Supervision, Writing - original draft, Writing - review & editing.

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Declaration of Competing Interest

The authors declared that there is no conflict of interest.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2020.117981>.

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